

# Sanjith Menon

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## Education

### **University of California, Los Angeles (UCLA)**

B.S., Mechanical Engineering, GPA: 3.958

Los Angeles, CA

Sep. 2022 – Jun. 2026

### **Stanford University**

M.S., Mechanical Engineering

Stanford, CA

Sep. 2026 – Jun. 2028

## Experience

### **Harbinger Motors – Product Engineering Intern, Powertrain**

*Jun. 2026 – Aug. 2026*

- Designed flywheel loading system supporting 1400+ lbs. to align and assemble inertial loads for park lock test rig.
- Analyzed fasteners in electric drive unit and recommended changes to patch lock and install torque for reliability.
- Fabricated oil temperature sensor harness for end-of-line coolant oil check to reduce faults in production units.
- Performed vibration (NVH) testing of oil pumps for electric drive unit and identified areas of interest for future test.

### **UCLA FSAE Bruin Formula Racing – Powertrain Engineer**

*Oct. 2022 – Jun. 2026*

- Designed ducting system for airflow enhancement and waterproofing of battery pack cooling and integrated it into powertrain system — achieved over 200% increase in air velocity during vehicle motion and passed rain test.
- Performed CFD analysis (Siemens) to validate cell segment temperatures and duct flow rates.
- Analyzed fan curves and cooling performance via MATLAB models based on heat transfer and fluid mechanics.
- Designed, manufactured, assembled motor test bench using machine shop skills to validate functionality of motor.

### **Autonomous Burger Assembly/Delivery Robot Capstone Project – Mechatronics Engineer**

*Jan. 2026 – Jun. 2026*

- Built and programmed an autonomous differential-drive robot to perform LiDAR-based course navigation, computer vision-guided customer recognition, and mechanical manipulation for burger assembly that ranked first in the capstone class.
- Implemented a non-blocking Python Finite State Machine (FSM) utilizing ROS 2 to orchestrate robot operation while prioritizing critical task timing and running a PID-based firmware control system.

### **Rivian Automotive – Manufacturing Engineering Intern, Paint**

*May. 2025 – Dec. 2025*

- Owned end-to-end design (CATIA), validation, and deployment of custom hood jig fixtures that eliminated 100% of bracket deformation and cut the number of distinct hood jigs in paint process by 25%.
- Led implementation, installation, and commissioning of \$200K e-coat air blower system reducing defects by 19% and of custom feather duster station reducing defects by 24%.
- Performed a data-driven root cause analysis of skid extension station faults and implemented and validated design changes, reducing faults by 99%.
- Developed proximity switch protection covers to minimize equipment damage and downtime — patent pending.

### **UCLA Structures-Computer Interaction Lab – Undergraduate Researcher**

*Feb. 2024 – Apr. 2025*

- Designed and fabricated end-effectors for experiments with LLM and vision-based control of cobot robotic arms.
- Programmed Python control functions for robotic arm manipulation for LLM library of tools.
- Performed rapid prototyping of sensor module combining NIR, FLIR, RGBD cameras for fruit detection project.
- Analyzed thermal data in MATLAB for shape-morphing composite material testing project.
- Co-authored research papers on shape-morphing composites (AEM, 2026) and on fruit detection via sensor fusion.

### **Southern California Edison – Mechanical Engineering Intern**

*Jun. 2024 – Sep. 2024*

- Developed an interactive tool for the Automated Demand Response program that reduced energy load shed calculation time from ~1 hour to 5 minutes, automated data visualization, and eliminated 90% of redundant processes in reporting, technical reviews, and data analytics.

## Skills

**Engineering Skills:** CAD (CATIA, SolidWorks), MATLAB, DFM, FMEA, V&V, FEA (Altair Inspire, Ansys, SolidWorks), CFD (Siemens), Breadboard/circuit design, Arduino, Sawyer Robot

**Programming and Tools:** Python, Java, JavaScript, HTML 5, C++, Microsoft Excel/Google Sheets, VBA, Microsoft Copilot, Power BI, Ubuntu, ROS2

**Machine Shop Skills:** 3D Printing, Laser Cutter, Mill, Lathe, Bandsaw, Drill Press, Dremel, Soldering